(a) Agreement with the labels, per system

Laguna del Maule
Nevados de Chillán
Copahue
Villarrica
Hudson
Ruiz-Olleta
Ruiz-Bis
Alaska
Cascade
Hawai`i
Northern California
Pavlof
Semisopochnoi
Veniaminof
Deception 2009

OVDAS
SGC
VCSEIS
VOISS-Net
Deception Is.

0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7

ARI with the catalogue labels

within-system, *FI–DET–SR*; median 0.22
within-system, 16 descriptors; median 0.11
joint, *FI–DET–SR* (k = 5); median 0.11
no stable partition (k = 2 reported)

(b) Joint grouping, *FI–DET–SR*, k = 5

		G0 high *FI* low *DET*	G1 high *SR*	G2 low *FI* high *DET* low *SR*	G3 low *DET*	G4 low *FI* high *DET*	
Laguna del Maule	LP			.80		.18	*OVDAS*
	VT		.58		.16	.21	
Nevados de Chillán	LP		.20		.21	.57	
	VT				.91		
	VLP	.10	.29		.53	.08	
	TO	.09	.56		.16	.18	
	AV	.06	.24		.66		
Copahue	LP		.12	.13	.09	.66	
	VT	.09	.28		.52	.08	
	HB		.44		.22	.34	
	VLP		.31		.16	.48	
Villarrica	LP		.25		.20	.51	
	VT	.88			.09		
	AV	.85			.12		
Hudson	LP		.05			.91	
	VT		.43		.31	.26	
	HB		.19		.06	.75	
	AV		.32		.20	.48	
Ruiz-Olleta	LP		.42	.11	.19	.27	*SGC*
	VT		.73		.13	.10	
	HB		.50	.05	.19	.26	
	TR		.18	.20	.10	.51	
Ruiz-Bis	LP		.72			.22	
	HB		.70	.07		.20	
	TR		.33	.14	.15	.36	
Alaska	LP		.24	.19		.55	*VCSEIS*
	VT		.60		.12	.26	
Regional			.61		.13	.19	
Other			.59		.25	.15	
Cascade	LP		.10			.88	
	VT		.64		.25	.11	
Hawai`i	LP		.16		.05	.79	
	VT		.54		.31	.15	
Northern California	LP		.11		.17	.72	
	VT		.75		.22		
Kilauea	VT		.54	.06	.12	.28	
Redoubt	VT		.45	.10	.14	.29	
Tanaga	VT		.47	.10	.15	.27	
Pavlof	LP	.20	.33		.21	.26	
TRh		.06	.11		.12	.71	
TRm		.05	.13		.13	.68	
TRb			.13		.06	.80	
Explosion			.20		.14	.64	
Noise		.13	.28		.53	.06	*VOISS-Net*
Semisopochnoi	LP		.24	.12	.17	.45	
TRh			.08		.08	.79	
TRm			.46		.21	.30	
TRb			.41		.24	.31	
Explosion		.09	.31		.28	.32	
Noise		.07	.13		.29	.51	
Veniaminof	LP		.27		.27	.41	
TRh			.23	.10	.08	.54	
Popocatépetl	TRh		.21		.22	.57	
Deception 2009	LP				.11	.85	*Deception Island*
NTR			.22		.29	.47	
Deception 1995	NTR		.16		.18	.63	

0.0 0.2 0.4 0.6 0.8 1.0

Fraction of the system–class in the group

(c) Stability of the joint grouping (large marker: chosen k)

Stability (median ARI between subsample fits)

1.0 0.9 0.8 0.7 0.6 0.5 0.4

0.8

2 3 4 5 6 7 8 9 10 11 12

Number of groups k

FI–DET–SR, all events (k = 5)
FI–DET–SR, class-balanced (k = 4)
FI–DET–SR, label-free sample (k = 3)
16 descriptors, all events (k = 2)
16 descriptors, class-balanced (k = 4)